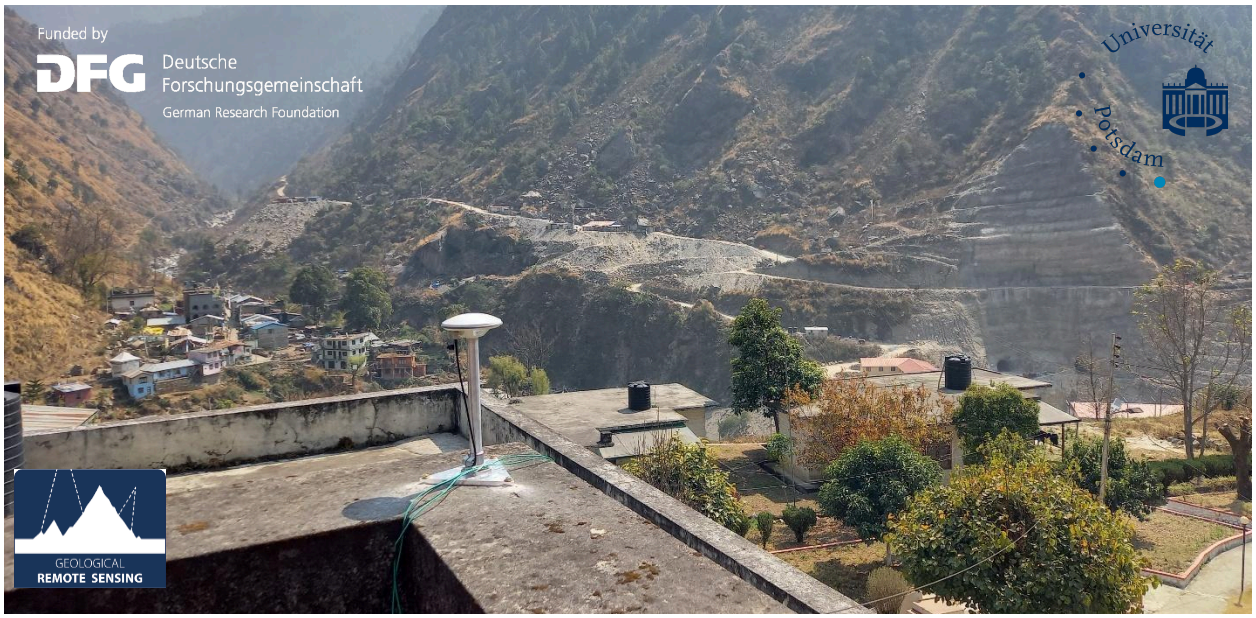
Combining Satellite and In-Situ Data for Ecosystem Monitoring

Falgun 10-11 (Feb 22-23), Department of Environmental Science | Kathmandu University

Led by Dr Taylor Smith (University of Potsdam, Germany), Dr Bodo Bookhagen (University of Potsdam, Germany), and Dr Shakil Regmi (South-Eastern Finland University of Applied Sciences, Finland). Research profiles are attached at the end of this document.



Workshop Goals: The workshop aims to present lessons learned from installing and maintaining an environmental sensor network in the Trishuli. The workshop will demonstrate how the sensors were built, installed, and what the data can be used for. There will be practical exercises introducing the data, as well as a strong focus on linking local environmental data to satellite time series. The participants will gain hands-on experience with open-source data processing tools, data visualization, and the design of open-source hardware.

Target Audience: The workshop is aimed at the meteorological and academic community, as well as those working in the areas of water management and disaster risk planning. We will cap the workshop at 18 participants so that everyone is able to follow the practical exercises.

Logistics: There are no fees to participate in this two-day workshop. **Each participant should bring their own laptop with a working Python environment for the hands-on exercises.** We will send out a document detailing computer setup instructions before the workshop.

The workshop will be organized at the **Kathmandu University (Dhulikhel Campus)**. Lunch, tea/coffee/cookies, and water will be provided during the workshop at the University.

Contact of Organizers (Shakil Regmi): +977 9841841974 | shakil.regmi@xamk.fi | Whatsapp: +49 15216782476

Registration:

Registration is open for 18 individuals on a first-come, first-served basis. The relevance of your background to the workshop theme will be considered. Please register for the workshop using the Google Form from this link: https://docs.google.com/forms/d/e/1FAIpQLSdJepHntk9LdlCK7tF9GNFav0POVejgZFNEq4rKkK2znMI20w/viewform?usp=publish-editor**.** The registration will be active until 1 Falgun (Feb 13), and we will confirm your spot for the workshop on 3 Falgun (Feb 15).

Details of Workshop Program

Day 1: Designing, building, and analysing in-situ sensor data in the Trishuli

09:30 – 10:00 Introduction from Kathmandu University

10:00 – 10:15 Collecting GNSS data in the Trishuli

10:15 – 10:30 Computer Setup

10:30 – 10:45 Tea/Coffee Break

10:45 – 11:15 Processing GNSS data to estimate water vapor changes

11:15 – 12:30 Practical Example – Working with GNSS RINEX data

12:30 – 13:30 Lunch Break

13:30 – 14:00 Low-cost environmental stations in the Trishuli

14:00 – 14:30 Introduction to using camera data for pixel tracking and river-velocity estimation

14:30 – 14:45 Tea/Coffee Break

14:45 – 15:45 Practical Example – Measuring river flow velocity with a camera

15:45 – 16:00 Wrap Up and Further Resources

Day 2: Contextualizing in-situ data with satellite remote sensing

09:30 – 10:00 Introduction to Google Earth Engine for data processing

10:00 – 10:30 Practical Exercise – Visualizing data using online tools

10:30 – 10:45 Tea/Coffee Break

10:45 – 11:15 Practical Exercise – Monitoring changes in high-elevation regions

11:15 – 12:30 Practical Example – Extracting Rainfall time series for specific locations

12:30 – 13:30 Lunch Break

13:30 – 14:30 Practical Example – Integrating watershed data and mapping climate trends

14:30 – 14:45 Tea/Coffee Break

14:45 – 15:45 Practical Example – Mapping changes in high-altitude rivers

15:45 – 16:00 Research Outlook, Ongoing Work, and Wrap Up

Brief Profile of Organizers

Dr. Taylor Smith is affiliated with the Institute of Geosciences at the University of Potsdam in Germany. His work aims to improve our understanding of earth-surface processes and the impacts of both short- and long-term climate changes using high-resolution data. He has worked for the last 10 years in the greater High Mountain Asia region, and now focuses more closely on the Nepalese and Indian Himalayas. His current projects explore the relationship between glaciers, snowmelt, and river hydrology. https://tasmi.github.io/

Prof. Dr. Bodo Bookhagen is a Full Professor at the Institute of Geosciences at the University of Potsdam in Germany. His work concentrates on hydrological and atmospheric processes within the fields of tectonics and geomorphology. His approach integrates remote sensing, field measurements, time series analysis, and numerical modeling. He has dedicated over 20 years to extensive research in the Nepalese and Indian Himalayas, as well as the South American Andes. https://bodobookhagen.github.io/

Dr Shakil Regmi is affiliated with the Department of Forestry and Environmental Technology at the South-Eastern Finland University of Applied Sciences in Finland. His work focuses on a comprehensive approach integrating environmental and earth observation of hydrometeorological processes with the realm of environmental psychometrics to understand the level of long-term environmental change and its impact on the environmental behaviour of people. His work is focused on the Nepalese and Central Himalayan regions.